TAMIL NADU STATE BOARD - STANDARD 9 SCIENCE (PHYSICS)

CHAPTER 2: MOTION

PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 9 English Medium
Subject:	Science (Physics)
Chapter Name:	Motion

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

TYPES OF MOTION – LINEAR, CIRCULAR, OSCILLATORY, RANDOM

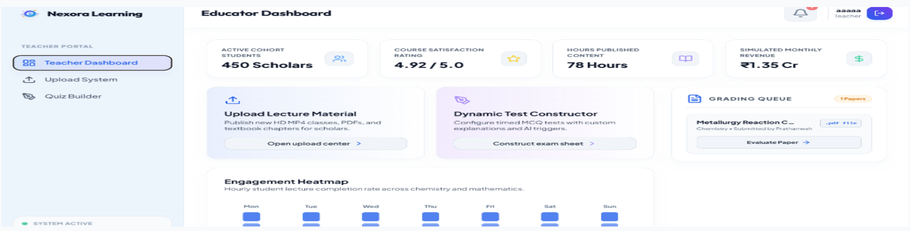
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Rest	An object is at rest if it does not change its position with respect to its surroundings. Example: A book on a table, walls of a room.
Motion	An object is in motion if it changes its position with respect to its surroundings. Example: Cars on a road, rotating fan.
Relativity of Motion	Motion is a relative phenomenon; an object can appear to be in motion to one observer and at rest to another depending on the frame of reference.
Distance	The actual length of the path travelled by a moving body, irrespective of direction. It is a scalar quantity (SI unit: m).
Displacement	The change in position of a moving body in a particular direction, representing the shortest distance between initial and final points. It is a vector quantity (SI unit: m).
Speed	The rate of change of distance or distance travelled per unit time. It is a scalar quantity (SI unit: m/s).
Velocity	The rate of change of displacement or displacement per unit time. It is a vector quantity (SI unit: m/s).
Acceleration	The rate of change of velocity; change in velocity per unit time. It is a vector quantity (SI unit: m/s^2).
Centripetal Force	The force causing centripetal acceleration, acting on a body in circular motion and directed radially towards the center ($F = mv^2/r$).
Centrifugal Force	An apparent outward force experienced by a body moving in a circular path, acting away from the center (fictitious/pseudo force arising from inertia).

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

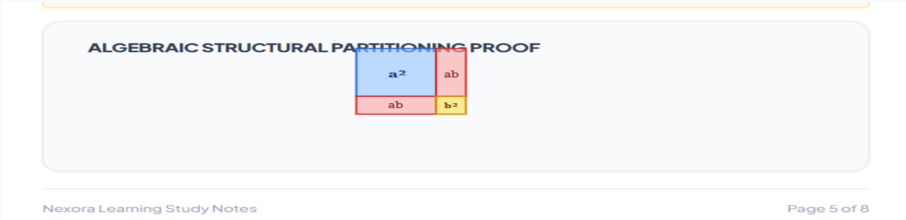
MOTION GRAPHS – DISTANCE-TIME & VELOCITY-TIME CURVES



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

EQUATIONS OF MOTION – V=U+AT, S=UT+0.5AT^2, V^2=U^2+2AS



Essential Formulas, Laws & Identities:

- Average Speed = Total Distance travelled / Total Time taken
- Velocity = Displacement / Time taken
- Acceleration: $a = (v - u) / t$
- First Equation of Motion: $v = u + at$
- Second Equation of Motion: $s = ut + (1/2)at^2$
- Third Equation of Motion: $v^2 = u^2 + 2as$
- Free Fall Equations (u=0): $v = gt$ | $s = (1/2)gt^2$ | $v^2 = 2gh$
- Uniform Circular Motion Speed: $v = (2 * \pi * r) / T$
- Centripetal Acceleration: $a = v^2/r$
- Centripetal Force: $F = m * a = (m * v^2) / r$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: A train starting from rest accelerates uniformly to a velocity of 36 km/h in 10 seconds. Find its acceleration in SI units.

Step-by-step Solution:

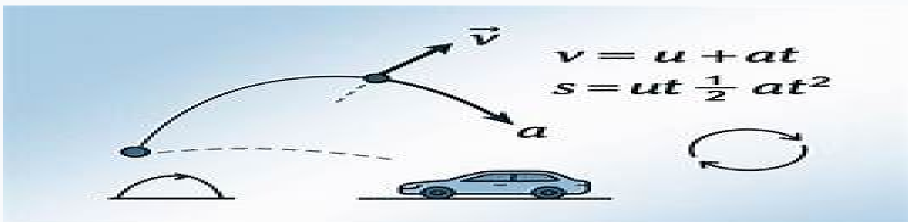
Convert velocity to m/s: $u = 0$ m/s, $v = 36$ km/h $= 36 * (5/18) = 10$ m/s. Time $t = 10$ s. Apply $v = u + at$: $10 = 0 + a*10 \Rightarrow a = 1$ m/s².

Example 2: A cyclist rides around a circular track of radius 40 m in 40 seconds. Calculate her speed.

Step-by-step Solution:

Given: $r = 40$ m, $T = 40$ s. Speed $v = (2 * \pi * r) / T = (2 * 3.1416 * 40) / 40 = 2 * 3.1416 = 6.28$ m/s.

LAWS OF MOTION – NEWTON'S 2ND & 3RD LAW ILLUSTRATIONS



HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Can an object have a constant speed but changing velocity? Give an example and explain.

Analytical Solution Strategy:
Yes. An object undergoing Uniform Circular Motion has a constant speed, but its velocity is constantly changing because its direction of motion is changing at every instant. This directional change constitutes an acceleration, directed toward the center (centripetal acceleration).

HOTS Challenge 2: If an object returns to its starting point after travelling a certain distance, what can you say about its distance and displacement?

Analytical Solution Strategy:
If the object returns to its starting point, its displacement is zero because the shortest distance between its initial and final positions is zero. However, the distance travelled is not zero, as it equals the actual length of the path traversed.

Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): Differentiate between distance and displacement with examples.

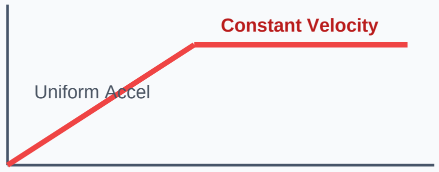
Q2 (2-Mark): Classify uniform and non-uniform motion and give real-life examples of each.

Q3 (2-Mark): Distinguish between speed and velocity.

Q4 (5-Mark): Deduce the three equations of motion from velocity-time graphs.

Q5 (5-Mark): Identify centripetal and centrifugal forces in daily life and state their directions.

PRACTICE: GRAPHING DISPLACEMENT VERSUS TIME



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Automotive Telemetry

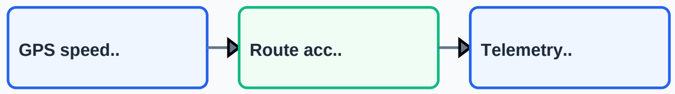
Speedometers measure instantaneous speed while GPS systems use velocity vectors to calculate real-time traffic and arrival times.

Application Case: Industrial Separation (Centrifugation)

Spin dryers in washing machines and blood separators in clinical labs utilize centrifugal force to separate liquid droplets/elements from solid matrices.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

APPLICATION: VEHICULAR SPEED AND TELEMETRY SYSTEMS



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.

✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.

✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3, N) for the final numeric answer.

✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

[ ] Read and memorized core definitions on Page 2.

[ ] Reviewed critical formulas and relations on Page 3.

[ ] Solved worked examples on Page 4.

[ ] Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress. Good luck!

REVISION: KINEMATICS MOTION SUMMARY FLOW